

CPE 381 CLASSWORK 01: INTRODUCTION TO SIGNALS AND SYSTEMS

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes

Due: August 24, 2026 (In-Class)

1. What is system thinking in Signals & Systems?

(5 Points)

Answer

In system thinking, a system is seen as an abstraction with a signal as an input and its modified version, modified by the system, as an output.

2. Consider a unit rectangular pulse in the top figure of Figure 1. Under what condition rectangular pulse tends to become an unit impulse signal? **(5 Points)**

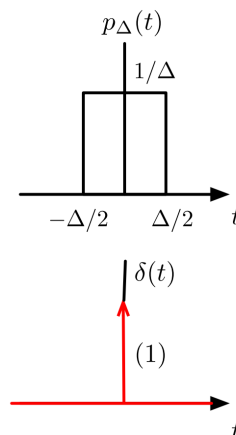


Figure 1: Rectangular Pulse and Unit Impulse Signal.

Answer

We consider that a unit rectangular pulse has area under the curve unity or 1. So, considering that, when $\Delta \rightarrow 0$, i.e. width of the pulse approaches zero, height approaches infinity, and unit rectangular pulse tends to become an unit impulse signal.

3. What does Linear Time-Invariant conditions says about a system? Explain in words. **(5 Points)**

Answer

In a linear time-invariant system,

- (a) if two signals provide some outputs when passing through a system, then adding those outputs to get combined output is same as as if the sum of signals was passed through a system to obtain the output.
- (b) When input is delayed or advanced by a certain amount, then output is equally delayed or advanced.

4. What's so special about δ -function, also known as unit impulse signal? **(5 Points)**

Answer

δ -function or unit impulse signals can be used to represent any signal as the weighted sum of shifted unit pulse signals. In other words, δ -function provides a generalized mathematical representation of any signal.